

Anurag Bheemani

Machine Learning Engineer | Computer Vision & Applied GenAI

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SUMMARY

Machine Learning Engineer focused on computer vision and video understanding. Hands-on with self-supervised video models (V-JEPA), object detection (YOLO), and model deployment via TensorRT and DeepStream, on a solid Python and backend foundation.

EXPERIENCE

Eizen AI - Machine Learning Engineer, Computer Vision | Hyderabad

Nov 2025 - Jul 2026

- Trained a lightweight attentive probe on Meta's frozen V-JEPA 2.1 encoder for 5-class bear-behavior recognition: 75.5% accuracy, 72.6% weighted F1, 66.9% clustering purity.
- Optimized a 375.5M-parameter V-JEPA 2 encoder's inference with PyTorch to ONNX to TensorRT/DeepStream (FP16): 28.9 ms vs 211.9 ms per clip (7.2x faster) and 39% less GPU memory; validated ONNX export across video model architectures.
- Improved leopard-detection accuracy from 83% to 92% using DINO-assisted auto-annotation, and deployed inference to NVIDIA Jetson edge devices.
- Automated labeling by dispatching 2,000-image batches every 10 minutes to a DINOv2/SAM auto-annotation workflow.
- Migrated Tesseract to PaddleOCR for system-monitoring screenshots, achieving 91% word accuracy across 150+ images.

GLOOMDEV - Web Developer Intern | Hyderabad

July 2024 - Dec 2024

- Built and deployed a MERN-stack e-commerce platform with secure APIs and database architecture, and integrated Razorpay payments, completing 50+ staging transactions.

SKILLS

Languages: Python, C++, JavaScript, SQL, TypeScript

ML & Deep Learning: PyTorch, scikit-learn, NumPy, pandas, Hugging Face Transformers, Vertex AI

Computer Vision: YOLOv8 (OBB), OpenCV, MediaPipe, SAM, DINOv2, CLIP, PaddleOCR, V-JEPA

LLM & GenAI Tools: Gemini SDK, LangChain, LangGraph, ChromaDB, RAG, Whisper, llama.cpp, whisper.cpp

Inference & Deployment: TensorRT, NVIDIA Triton, NVIDIA DeepStream SDK, ONNX, NVDEC, Docker, CUDA, Vulkan

Simulation & Edge: NVIDIA Isaac Sim, NVIDIA Jetson, Raspberry Pi (Hailo accelerator)

Backend, Data & Desktop: FastAPI, Node.js, REST APIs, Kafka, MongoDB, MySQL, PostgreSQL, Redis, GCP Cloud Run, Tauri

Tools: Git, Linux

PROJECTS

Paper Tail - Agentic Research Assistant | [Link](#)

Python, LangGraph, LangChain, FastAPI, FAISS, React

- Built Paper Tail, an agentic RAG research assistant for arXiv papers, with an 8-node LangGraph workflow: query classification, retrieval planning, evidence evaluation, iterative decisions, citation traversal, claim verification, and report generation.
- Implemented agent-controlled retrieval picking between 2 strategies per query (semantic, BM25) plus automatic year-based filtering, with citation-graph traversal for a paper's own citations when deeper source context is needed.
- Designed an iterative evidence loop (capped at 3 iterations) evaluating evidence for relevance, gaps, and conflicts, with final claims checked against their supporting sources.

Orion - Offline AI Video Player | [Link](#)

Tauri, C++, llama.cpp, whisper.cpp, FastAPI

- Built a commercial offline AI desktop video player for Windows with speech transcription, translation into 14+ languages, semantic search, and AI chat.
- Integrated whisper.cpp and llama.cpp with CUDA, Vulkan, and CPU backends, delivering up to 5-10x faster local inference than CPU-only execution.
- Engineered a fully offline C++ RAG pipeline with custom ONNX embeddings, binary vector indexing, and persistent model serving to eliminate per-request loading overhead.
- Built the backend (FastAPI, PostgreSQL, GCP Cloud Run) for authentication, licensing, model delivery, and payment processing (Razorpay, Paddle) with signature-verified webhooks.

Face Sorter - Photo Organizer | [Link](#)

Python, FastAPI, React Native, MongoDB

- Developed a React Native + FastAPI app that groups photos by face using the `face_recognition` library (dlib 128-D encodings, Euclidean-distance matching), stored in MongoDB.
- Grouped 100+ photos per batch at about 70% correct grouping, removing manual sorting.

EDUCATION

Jawaharlal Nehru Technological University Hyderabad

B.Tech, Computer Science (AI & Machine Learning)

Dec 2021 - Jul 2025

CGPA: 6.98/10.00

ACHIEVEMENTS

- "Exceptional Intern of the Month" during web-development internship at GLOOMDEV.
- Led 24 members across 6 groups for the technical execution of the college annual tech fest.

CERTIFICATION

- Web Development Internship – [GLOOM DEV](#)
- Edunet Code Unnati Advance Course – [EDUNET](#)
- Code Unnati Innovation Marathon 2025 – [EDUNET](#)